Pro Football Sports Science Research Update

February 2026

A comparison of brain MRI outcomes in youth American football versus non-contact sport athletes. Medicine and Science in Sports and Exercise, 58(2): 261-270. 2026.

A systematic review and meta-analysis of anterior cruciate ligament injury prevention programs: Training to stay in the game. The American Journal of Sports Medicine, Ahead of Print: 1-8, 2026.

Analyzing ankle injuries in the NFL: A video-based investigation of the impact of artificial turf on injury mechanisms. Journal of Orthopaedic Experience & Innovation, 7(1): 1-7, 2026.

Athlete perceptions of in-person training, extended reality and video technologies for perceptual-cognitive development: A qualitative exploration of developmental American football quarterbacks' perspectives. International Journal of Sports Science & Coaching, Ahead of Print: 1-16, 2026.

Big bones mean big muscles: An MRI-based dataset of muscle-bone-body size relationships across 70 human muscles of the upper limb, trunk, and lower limb. Journal of Applied Physiology, 140: 1-19, 2026.

Biomechanical impact of soccer shoe outsole design on knee joint during cutting movements. Footwear Science, 17: S264-S265, 2025.

Competitive position, expectations, and managerial approaches to risk in National Football League football. Journal of Sport Management, 39: 342-353, 2025.

Does timing of concussion in-season affect incidence and recovery in Division 1 collegiate football players? A CARE Consortium Study. Journal of Head Trauma Rehabilitation, Ahead of Print: 1-10, 2026.

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Effect of high vs low volume of the Nordic hamstring curl on hamstring muscle architecture and eccentric strength in soccer players: A systematic review and meta-analysis. Frontiers in Physiology, 16: 1631205, 2025.

Effects of resisted-sprint training on sprint performance and mechanics: A systematic review and meta-analysis focusing on load magnitude. Scandinavian Journal of Medicine & Science in Sports, 25: e70182, 2025.

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The use of fantasy points to evaluate return-to-play performance after time-loss injuries in the National Football League. Cureus, 17(12): e99333, 2025.

Transdermal uptake of substances banned in sports and its relevance for doping controls. International Journal of Sports Medicine, Ahead of Print: 1-19, 2025.

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A comparison of brain MRI outcomes in youth American football versus non-contact sport athletes. Medicine and Science in Sports and Exercise, 58(2): 261-270. 2026.

Objectives: This study aimed to compare brain magnetic resonance imaging (MRI) outcomes between children who play American football versus noncontact sport controls testing the hypotheses that history (primary) and duration (secondary) of football participation would be associated with differences in cortical thickness, subcortical volume, resting-state functional connectivity, and white matter diffusivity.

Methods: This secondary analysis of cross-sectional baseline data from the Adolescent Brain Cognitive Development Study compared brain MRI outcomes between 9- and 10-year-old children who play American football (n = 1194) versus noncontact sport controls (n = 807). Outcomes included 74 bilateral cortical thickness regions; 10 gray matter subcortical volumes, with a priori focus on the hippocampi; resting-state functional connectivity (169 network-network correlations and 247 network-region correlations across 13 resting-state functional networks and 19 regions); and 21 diffusion tensor measures.

Results: Football participation was associated with global effects on cortical thickness (P = 0.017), network-to-network resting-state connectivity (P = 0.010), and fiber tract volume (false-discovery rate [FDR]-adjusted P = 0.015) in primary analysis, but the only significant post hoc finding after FDR correction was smaller cortical thickness adjacent to the left anterior transverse collateral sulcus in the football group (Cohen d = -0.258, FDR-adjusted P = 0.017). There were no significant duration of football play effects in secondary analyses (all P > 0.05). Targeted analysis of hippocampal volumes yielded no significant football or duration of play results (both P > 0.05), but suggested a potential trend of lower hippocampal volumes with increasing duration of play.

Conclusions: At ages 9-10, participation in American football was associated with minimal differences across a large array of structural, functional, and diffusion tensor MRI outcomes. While the clinical implications of these cross-sectional results are unknown, they merit additional investigation and can contribute to the ongoing discussion surrounding contact sport participation in children.

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A systematic review and meta-analysis of anterior cruciate ligament injury prevention programs: Training to stay in the game. The American Journal of Sports Medicine, Ahead of Print: 1-8, 2026.

Background: Approximately 400,000 anterior cruciate ligament (ACL) reconstructions are performed each year in the United States. Effective ACL injury prevention programs may be paramount in reducing this significant injury burden.

Purpose: To determine the effectiveness of ACL injury prevention programs and generate updated guidelines to protect athletes from these injuries.

Study design: Meta-analysis, Level of evidence, 2.

Methods: The Embase, PubMed, and Ovid (MEDLINE) databases were searched in accordance with PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. Included studies focused on ACL injury prevention as opposed to treatment and provided data on ACL injury rates after intervention implementation. Using random-effects models, the authors generated the pooled risk ratio (RR) for all data and female-only data, <18 years versus ≥18 years of age data, handball data, soccer data, and balance board data.

Results: Eighteen articles were identified (9 randomized controlled trials, 9 prospective cohort studies). The 25,166 studied athletes (mean age, 19.3 ± 3.6 years; >85% female) played handball, soccer, basketball, or volleyball. All interventions were studied for a minimum of 1 season (mean, 1.3 ± 0.59). Athletes who participated in an ACL injury prevention program were significantly less likely to sustain an ACL rupture with a pooled RR of 0.46 (95% CI, 0.36-0.57). When analyzed by age, there was a significant risk reduction in ACL rupture for both athletes <18 years and ≥18 years (RR, 0.35 [95% CI, 0.22-0.55] and RR, 0.50 [95% CI, 0.38-0.64], respectively). The pooled RR was also statistically significant for female players (RR, 0.57 [95% CI, 0.43-0.74]), soccer and handball athletes (RR, 0.30 [95% CI, 0.19-0.46] and RR, 0.66 [95% CI, 0.46-0.96], respectively), and players participating in programs including balance boards (RR, 0.49 [95% CI, 0.35-0.67]).

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Conclusion: Athletes who did not partake in an ACL injury prevention program were nearly twice as likely to sustain an ACL rupture compared with those who did. This study provides strong support for using neuromuscular training programs to significantly reduce the risk of ACL rupture among athletes.

Analyzing ankle injuries in the NFL: A video-based investigation of the impact of artificial turf on injury mechanisms. Journal of Orthopaedic Experience & Innovation, 7(1): 1-7, 2026.

Purpose: Artificial turf is increasingly used in professional football due to its economic and logistical advantages. However, concerns persist that certain turf types, especially slit-film, may alter biomechanics and increase the risk of lower extremity injuries. This study aimed to evaluate the mechanism and frequency of ankle and foot injuries across three surface types, natural grass, monofilament/dual-fiber (MFDF) turf, and slit-film turf, during the 2022–2023 NFL season. We hypothesize that slit-film turf will show a higher rate of ankle and foot injuries compared to other types of turf and natural grass.

Methods: A retrospective review of NFL injury reports and game footage was conducted to identify 78 in-game ankle and foot injuries. Injuries were classified by mechanism (contact vs. non-contact) and correlated with surface type. Statistical analyses including Mann-Whitney U, Chi-Square, and logistic regression were employed to assess associations.

Results: Slit-film turf exhibited the highest rate of contact injuries (85.7%) but the lowest rate of non-contact injuries (14.3%), contrary to the hypothesis and prior research. Natural grass had the highest proportion of non-contact injuries (46.4%). Statistically significant differences were found between surface type and injury mechanism ($p = 0.015$). Fatigue may contribute to injury prevalence, as the fourth quarter saw the highest injury rate (41%).

Conclusion: This study challenges the prevailing belief that artificial turf, particularly slit-film turf, increases non-contact injury risk. Instead, slit-film turf may predispose players to more contact-related injuries due to its rigid surface properties. Findings underscore the need for multi-season, biomechanically informed research to guide player safety policies.

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Athlete perceptions of in-person training, extended reality and video technologies for perceptual-cognitive development: A qualitative exploration of developmental American football quarterbacks' perspectives. International Journal of Sports Science & Coaching, Ahead of Print: 1-16, 2026.

Abstract - Advancements in technologies such as 360°video and virtual reality (VR) open new possibilities for how athletes can train their perceptual-cognitive skills. However, few studies have explored the athletes' perception of these different training approaches qualitatively. Therefore, this study engaged three quarterbacks (QBs) from an Australian American football team in four different sport-specific training approaches (on-field, 360°video, VR, traditional video) and explored their thoughts and preferences in relation to how each could help develop their perceptual-cognitive skills. The developmental QBs participated in semi-structured interviews after each of the training approaches. Interpretive Phenomenological Analysis was used to explore participants' unique experiences of each approach. Two primary themes were identified, along with secondary themes: 'Opportunities for training perceptual-cognitive skills' (sub-themes: Making the best decision; Where to look; Awareness and reflection), and 'Factors influencing perceptual-cognitive skill development' (subthemes: Control of movements; Realism). All participants ranked on-field training as their preferred approach, with two participants then ranking the technologies as 360°video > VR > traditional video, and one participant ordering them VR > traditional video>360°video. The findings can be contextualized through an ecological dynamics framework; specifically, the first theme in relation to affordances for action, while the second theme relates to perception-action coupling and representative learning design. Overall, the study provides unique individual insights on QBs' perceptions and preferences towards technologies such as 360°video and VR and how they may be utilized to develop perceptual cognitive skills. Future research should look to build on these findings by examining whether the positive perceptions transfer to actual performance improvements in a longitudinal intervention design.

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Big bones mean big muscles: An MRI-based dataset of muscle-bone-body size relationships across 70 human muscles of the upper limb, trunk, and lower limb. Journal of Applied Physiology, 140: 1-19, 2026.

Abstract - Body sizes and shapes vary widely, even among healthy adults, resulting in diverse muscle sizes, strengths, and performance capacities. This study developed an artificially intelligent (AI) algorithm to segment individual muscles and bones from whole body MRI scans of 102 healthy adults (49 males, 53 females) aged 18-50 yr, generating three-dimensional (3-D) segmentations of 70 muscles and 13 bones spanning the upper limbs, trunk, and lower limbs. We quantified muscle volume, asymmetry, and fat fraction at whole body, regional, and individual-muscle levels, and examined how these properties correlate with body size and skeletal dimensions. Fat fraction and asymmetry varied across muscles and were generally similar between sexes; however, the distribution of muscle volume across the body differed between females and males. Across all predictors tested, total bone volume showed the strongest correlation with total muscle volume (r^2 = 0.85), followed by femur volume, height × mass, mass, height, and BMI. At the individual muscle level, the associated bone volume consistently explained more variance in muscle size than anthropometric predictors. Correlations between muscle volume and body-size parameters were significantly different between males and females, whereas bone-volume correlations showed no significant sex differences. These results suggest that skeletal dimensions-reflecting an individual's "frame size"-are stronger determinants of muscularity than body size metrics and explain the observed sex differences in muscle sizes. This work presents a comprehensive in vivo muscle-level dataset to date, introduces a novel framework for analyzing muscle-bone correlations, and provides reference data for applications from clinical diagnostics to athletic performance and musculoskeletal modeling.

NEW & NOTEWORTHY - This study presents the most comprehensive in vivo dataset of full-body muscle and bone volumes in healthy adults, showing that skeletal dimensions are the strongest predictors of muscularity, with height × mass emerging as the second-best predictor.

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Biomechanical impact of soccer shoe outsole design on knee joint during cutting movements. Footwear Science, 17: S264-S265, 2025.

This study examines how variations in soccer cleat outsole design—specifically forefoot stud number and distribution—affect knee biomechanics during sidestep cutting maneuvers. While focused on soccer, the findings have direct parallels to American football, where cleat traction influences ACL injury risk during similar non-contact cutting and pivoting actions common in NFL play (e.g., route running, evasive maneuvers). The research underscores the trade-off between enhanced performance (via better traction) and elevated joint loads, which could inform cleat selection, training protocols, and injury mitigation strategies in sports medicine and performance coaching.

Key Study Design and Methods

- **Participants**: 17 healthy male collegiate-level soccer players (no NFL equivalents specified, but comparable to athletic males in high-demand pivoting sports).
- **Conditions**: Two otherwise identical cleats tested in randomized order:
  - ATM1: 8 studs in the forefoot.
  - ATM2: 9 studs in the forefoot (increased number and redistributed for potentially better pressure dispersion).
- **Task**: 45° sidestep cutting with ball dribbling, simulating game-like direction changes. This mirrors NFL scenarios like receiver cuts or defensive back pivots, where 6-9% of ACL injuries occur during similar movements (citing Patel et al., 2023).
- **Data Collection**: Kinematics via 12 Vicon cameras (250 Hz); kinetics via Kistler force plate (1000 Hz). Markers on lower limbs; data processed in Visual3D with a 6-DOF model. Moments normalized to body weight ($N \cdot m \cdot BW^{-1}$).
- **Analysis**: Paired t-tests ($p < 0.05$) for differences between conditions.

Main Findings

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No significant differences in peak vertical ground reaction force (PVGRF), knee varus moment, external rotation moment, flexion moment, or extension moment. However, key variables showed statistically significant increases with the 9-stud design (ATM2), suggesting higher traction comes at a biomechanical cost:

Variable	ATM1 (8 Studs)	ATM2 (9 Studs)	p-value	Implication for NFL Context
Dynamic Traction Coefficient (DCT)	0.27 (0.03)	0.34 (0.02)	<0.05	Higher traction improves stability and acceleration but may "stick" the foot more, increasing torque transfer to the knee—relevant for turf vs. grass cleats in football.
Knee Valgus Moment ($N \cdot m \cdot BW^{-1}$)	0.83 (0.32)	0.98 (0.27)	<0.05	18% increase; excessive valgus loading is a known ACL risk factor in cutting sports, potentially exacerbating medial knee stress in NFL athletes with high cleat grip.
Knee Internal Rotation Moment ($N \cdot m \cdot BW^{-1}$)	0.79 (0.27)	1.16 (0.56)	<0.05	47% increase; combined with valgus, this elevates rotational stress, mirroring mechanisms in football ACL tears during planted cuts.

These changes indicate that denser stud configurations enhance grip (higher DCT) but amplify knee valgus and internal rotation moments, which are linked to ACL strain (per references like O'Connor & James, 2013). The study notes no differences in other planes (e.g., sagittal), but the coronal and transverse plane effects could compound over repetitive cuts in a game (thousands per match in soccer; analogous to NFL drives).

Discussion and Implications for NFL Sports Medicine/Performance

The authors conclude that while more forefoot studs improve traction and stability—potentially boosting cutting performance—they may heighten knee injury risk by increasing joint pressure and rotational forces. This aligns with broader literature on footwear-surface interactions (e.g., Palucci Vieira et al., 2024), where excessive grip without adequate release can overload the knee.

For NFL professionals:

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- **Injury Prevention**: Consider cleat stud density when prescribing footwear, especially on high-friction surfaces like modern turf. Athletes with ACL history or valgus-prone knees might benefit from lower-stud designs to reduce moments, balanced against performance needs.
- **Performance Optimization**: Higher traction could enhance agility metrics (e.g., 5-10-5 shuttle times), but monitor for cumulative load via wearables or biomechanical assessments. Integrate with training: Emphasize neuromuscular control (e.g., landing mechanics) to mitigate risks.
- **Limitations and Future Directions**: Small sample (n=17 males only); no long-term injury tracking; focused on knee (though purpose mentions "ankle" possibly in error—data is knee-specific). NFL-relevant extensions could test football-specific cleats (e.g., molded vs. detachable studs) or incorporate variable surfaces/weather.
- **No Conflicts**: Authors report none.

Overall, this study highlights the need for personalized cleat selection in cutting-intensive sports like football, weighing traction gains against biomechanical risks to optimize player health and longevity. If adapting to NFL protocols, cross-reference with football-specific data (e.g., from NFL Combine testing or injury registries).

Competitive position, expectations, and managerial approaches to risk in National Football League football. Journal of Sport Management, 39: 342-353, 2025.

Abstract - This paper analyzes how National Football League coaches incorporate information on competitive position and expectations of winning into fourth down decisions. Theoretically, teams should take more risk when their win probability is relatively low since additional risk broadens the distribution of possible outcomes thereby increasing the likelihood of victory. Our findings confirm this. We find that heavier favorites and teams having had higher levels of win probability within the game attempt more fourth down conversions, both on plays where attempting a play is estimated to be optimal and on plays where it is not. One possible explanation for this is coaches in those positions more likely frame their current win probability as being below a previously defined reference point, which

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reduces their degree of loss aversion. On balance, we find that this increased aggressiveness on fourth down improves the expected changes in win probability in this context.

Does timing of concussion in-season affect incidence and recovery in Division 1 collegiate football players? A CARE Consortium Study. Journal of Head Trauma Rehabilitation, Ahead of Print: 1-10, 2026.

Objective: To investigate whether seasonality is associated with (1) concussion incidence and (2) recovery outcomes, including the time to symptom resolution (SR) and return-to-play (RTP) in Division I college football players.

Setting: Multi-institution study using the CARE Consortium database composed of 30 universities investigating the natural history of concussions between 2014 and 2020.

Participants: 682 Division I football athletes across 21 institutions who sustained a concussion during practice or competition.

Design: Prospective cohort study of NCAA Division I football players with concussion.

Main measures: Concussion incidence rates across 4 season periods (preseason, early, middle, and late-season) were analyzed using chi-square tests. Time to SR and RTP were analyzed using Kaplan-Meier curves, and Cox proportional hazards regression.

Results: Concussion incidence rates were highest during preseason (4.73 per 100 players) compared to early (2.19), middle (2.31), and late-season (2.17). The temporal distribution showed 41.9% of concussions occurred during preseason, with 19.2%, 20.1%, and 18.8% occurring during early, middle, and late-season, respectively ($\chi2 = 104.6$, $P < .001$). Kaplan-Meier analysis revealed significant differences in time to SR (log-rank $P = .012$). Early ($HR = 0.582$, $P < .001$) and late-season ($HR = 0.634$, $P = .002$) concussions had delayed SR compared to preseason. No differences in time to RTP (log-rank $P = .300$) were found, although early-season concussions had a lower RTP hazard ($HR = 0.703$, $P = .040$).

Conclusions: In a multi-institution cohort of collegiate football players, preseason resulted in the highest incidence of concussions, while early and late-season concussions had significantly longer time to SR

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compared to preseason. These temporal patterns suggest that contextual factors may influence concussion outcomes and warrant mechanistic investigation to inform future management strategies.

Effect of high vs low volume of the Nordic hamstring curl on hamstring muscle architecture and eccentric strength in soccer players: A systematic review and meta-analysis. Frontiers in Physiology, 16: 1631205, 2025.

Abstract - Hamstring strain injuries (HSI) remain a significant problem in professional soccer, as this injury is the most prevalent. Nordic hamstring exercise (NHE) is the most researched exercise regarding its effect on modifiable factors of HSI. However, there is still debate about the minimal effective dosage for this exercise. This systematic review and meta-analysis aimed (1) to analyse the effects of low- and high-volume NHE on eccentric strength and hamstring muscle architecture in soccer players, and (2) identify gaps in the literature to guide future research. Three electronic databases (PubMed, Web of Science, Scopus) were searched, and 11 studies met the inclusion criteria. Pooled effect sizes (Hedges' g) and 95% confidence intervals (CI) were calculated using a random-effects model. High-volume NHE interventions significantly improved eccentric hamstring strength ($g = 0.77$, 95% CI 0.49-1.06, $p < 0.001$, $I^2 = 51\%$), fascicle length ($g = 0.43$, 95% CI 0.20-0.65, $p < 0.001$, $I^2 = 0\%$), and muscle thickness ($g = 0.48$, 95% CI 0.28-0.68, $p < 0.001$, $I^2 = 0\%$). Effects on pennation angle were non-significant ($g = -0.16$, 95% CI -0.38-0.06, $p = 0.16$). Low-volume protocols significantly increased eccentric strength ($g = 0.46$, 95% CI 0.06-0.87, $p < 0.05$, $I^2 = 0\%$) but did not result in meaningful changes in fascicle length, pennation angle, or muscle thickness. For eccentric torque, neither high or low volume interventions produced significant effects (both $g \approx 0.04$, $p = 0.74$, $I^2 = 0\%$). Control groups across all outcomes showed only trivial or negative changes. Results indicate that high volume of NHE (~2-3 sets of 8-12 repetitions/2-3 times per week) significantly increases peak eccentric strength, fascicle length of biceps femoris long head, and muscle thickness, while pennation angle shows only trivial increase. Low volume of NHE (~1-2 sets of 3-5 repetitions/1-2 times per week) shows a similar effect on peak eccentric hamstring strength, but there are no improvements in hamstring muscle architecture. Additionally, the effect of both types of volume on hamstring eccentric peak torque seems to be inconsistent. The variability of different testing methods on isokinetic strength and small correlations between other methods introduce challenges in comparisons with eccentric strength outcomes. A high volume of NHE seems to influence hamstring

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architecture adaptations better than low volume despite no differences in eccentric peak strength. Factors such as the player's different level (amateur, semi-professional or professional), previous experience with NHE, and compliance significantly influence the training outcomes. Future research is needed to better determine the effect of low volume of NHE on the hamstring architecture adaptations in soccer players regarding previous experience with NHE and playing level. Furthermore, standardization of assessment tools and outcome measures is critical for future comparisons with isokinetic dynamometry.

Effects of resisted-sprint training on sprint performance and mechanics: A systematic review and meta-analysis focusing on load magnitude. Scandinavian Journal of Medicine & Science in Sports, 25: e70182, 2025.

Abstract - Resisted sprint training (RST) is effective for improving sprint acceleration. However, the choice of RST loads-ranging from light (≤ 10% of unresisted sprint maximum velocity decrement [v_{dec}], < 20% of body mass [BM]) to very heavy (≥ 50% v_{dec} or ≥ 80% BM) -remains debated. To (i) explore the effects of different RST loads and unresisted sprint training (UST) on sprint performance and macroscopic mechanical outputs (e.g., theoretical maximal horizontal force [F_0], power [P_{max}], maximal ratio of horizontal-to-resultant force [RF_{max}]), and (ii) investigate how participant characteristics and training parameters influence sprint performance at varying RST loads. A search of 10 databases was conducted until September 2024, with studies included if they involved longitudinal interventions using horizontal resistance (e.g., sled) and if RST was the primary difference in training between groups. Analyses included single-arm (RST pre vs. post, considered exploratory only), two-arm (RST vs. UST), and multi-arm comparisons (different RST loads, UST, and control). Analytical models included two-level, three-level, and Bayesian-based network meta-analysis. The effects of different moderators on sprint performance and mechanical characteristics were explored through subgroup analyses and meta-regressions. Certainty of evidence was assessed using the GRADE approach. A total of 49 RST studies (1281 participants: 1153 males, 160 females, 64 unclear) were included. Pooled results showed RST was more effective than UST for improving sprint performance (effect size [ES] = -0.30, Hedges' g, Moderate GRADE). However, only moderate (≥ 20%-50% BM or > 10%-30% v_{dec}) and very heavy RST were more effective than UST, whereas light and heavy loads (> 50%-80% BM or > 30%-50% v_{dec}) were not. Two-

and multiple-comparison models indicated that very heavy RST loads produced significantly greater performance improvements than light loads. The effect of RST on performance compared to UST became more pronounced with an increasing load ($\beta = -0.005$, $p = 0.018$), with a significant threshold at 20.7% BM, beyond which RST led to significantly greater performance improvements than UST ($p < 0.05$). Very heavy RST loads were particularly effective in improving F_0, P_{max}, and RF_{max} compared to UST, with adaptations in F_0 increasing significantly in tandem with RST ($\beta = 0.010$, $p = 0.023$). These effects were more pronounced in trained and highly trained individuals, with negligible improvements in recreationally active participants. There was an inverted U-shaped relationship between training volume (total distance) and training effect: heavy loads at small volumes yielded similar benefits to light loads at large volumes. Finally, participant characteristics (age, sex) and training parameters (duration, frequency, number of training sessions, sprint phase, session volume) also influenced sprint performance improvements. Very heavy loads were generally more effective at increasing sprint performance than UST and light RST loads, with light RST showing comparable effects to UST. These findings were likely attributable to improvements in the early acceleration phase (especially 0-5 m). Heavier loads were optimal for trained and highly trained groups, but no clear benefit of higher loads for recreational groups, suggesting light RST or UST are probably sufficient. Coaches and practitioners can refine protocols based on these findings to boost sprint performance outcomes.

Electrocardiographic diversity in football players: How distinct ethnic patterns shape athletes' hearts. European Journal of Preventive Cardiology, 33: 6-7, 2026.

This article, published in the *European Journal of Preventive Cardiology* (2026), features an interview with Brazilian sports cardiology researchers Filipe Ferrari and Ricardo Stein, based on their four studies analyzing resting ECGs from over 6,000 male Brazilian soccer players (ages 15-35, ethnically diverse: White, Black, Mixed-race). The work highlights how ethnic admixture influences ECG interpretation in athletes, challenging one-size-fits-all guidelines and offering insights relevant to NFL preparticipation screening, where diverse player populations (e.g., African American, mixed-ethnicity) are common. Key takeaways tailored for NFL sports medicine and performance pros:

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Study Overview and Methods

- **Dataset**: Longitudinal data (up to 20 years) from elite and sub-elite Brazilian soccer clubs, including ECGs, echocardiography, and follow-up (mean 4 years). No sudden cardiac deaths reported.
- **Ethnic Focus**: Brazil's high genetic admixture (e.g., African, European, Indigenous roots) dilutes "pure" ethnic ECG patterns, making findings applicable to multicultural teams like NFL rosters.
- **Comparison to Norms**: Builds on international criteria (e.g., Drezner et al., 2017), but addresses gaps in mixed-race athletes, who are underrepresented globally.

Key ECG Findings

- **Overall Normality**: 97% of ECGs were normal; only 3% abnormal—lower than in homogeneous African cohorts (23% abnormalities). Abnormalities increased with age (peaking 26-35 years), suggesting age-stratified screening could optimize NFL protocols.
- **Ethnic Variations**:
    - **Black Athletes**: Higher prevalence of training-related changes (e.g., early repolarization in 48%, longer PR intervals, left ventricular hypertrophy criteria). However, the classic "African/Afro-Caribbean" pattern (e.g., convex ST elevation) was rare (~1-2%), likely due to admixture—contrasting with purer African or Afro-Caribbean groups.
    - **Mixed-Race Athletes**: Intermediate patterns between White and Black (e.g., 41% early repolarization). These findings were benign, with no echo abnormalities, emphasizing the need for admixture-aware guidelines to reduce false positives in diverse NFL players.
    - **White Athletes**: Lower rates of changes (e.g., 30% early repolarization).
- **Early Repolarization (ER)**: Common (40% overall), classified into 7 morphologies. Most (e.g., J-wave with ascending ST) are benign and training-related. Riskier patterns (e.g., horizontal ST or inferolateral J-waves) were rare and not linked to adverse outcomes.
- **Other Patterns**: Inferolateral T-wave inversions warrant further imaging (e.g., cardiac MRI), even if echo is normal, as they may signal pathology.

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Implications for NFL Screening and Performance

- **Reinforces Current Practices**: International ECG criteria effectively minimize false positives while identifying risks in admixed populations. In NFL contexts, this supports tailored interpretations to avoid over-testing young, diverse athletes, potentially reducing downtime and costs.
- **Challenges and Recommendations**:
    - Stop over-worrying about benign ER—it's common and not associated with SCD in athletes.
    - Never overlook inferolateral T-wave inversions; pursue advanced imaging.
    - Advocate for updated guidelines to include mixed-race data, as Brazil's admixture mirrors U.S. diversity (e.g., higher false positives in African American players if using non-admixed norms).
- **Future Directions**: Researchers are expanding to female athletes (n>1,200), anticipating sex differences (e.g., more V1-V3 T-wave inversions, less ER/LVH). This could inform NFL women's programs or co-ed training insights.
- **Performance Tie-In**: Understanding ethnic ECG diversity aids in distinguishing physiologic adaptations (e.g., "athlete's heart") from pathology, enabling safer training intensification and personalized recovery protocols in high-admixture teams.

This research underscores precision in ECG screening for ethnically diverse athletes, potentially reducing unnecessary referrals while enhancing detection of true risks—directly translatable to NFL workflows for optimizing player health and longevity. References to the four studies and supporting literature are included in the original for deeper dives.

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Evidence-based management of patellar tendinopathy in athletes: A narrative review. Current Physical Medicine and Rehabilitation Reports, 14: 4, 2026.

Purpose of Review: To synthesize the available literature for evidence-based management of patellar tendinopathy (PT) among athletes and to provide a conceptual framework to help guide clinical decision-making when managing in-season and off-season athletes with PT.

Recent Findings: Physical therapy remains the first line treatment for PT. Several studies have reported no improvement or inferior outcomes in patients treated with corticosteroid injection. Preliminary studies show promising outcomes for athletes treated with tendon scraping and percutaneous ultrasonic tenotomy. Recent studies report high rates of return to sport with arthroscopic and open surgical treatments in refractory cases.

Summary: Despite PT being a common condition affecting athletes, the optimal management remains undetermined. The location and severity of patellar tendon pathology, individual characteristics of the affected athlete, timeline of the athletic season, recovery time post-intervention, treatment efficacy, associated cost, and potential treatment side effects should be considered when determining management. Multiple effective and safe non-operative treatment options are available to manage this condition with surgery available for refractory cases. The treatment framework described here is based on a comprehensive review of the available evidence.

Exploring sprint kinematics in American football athletes after anterior cruciate ligament reconstruction. Scandinavian Journal of Medicine & Science in Sports, 36: e70200, 2026.

Abstract - The purpose of this study was to compare stance time and knee joint kinematics between limbs during sprinting in Division-1 collegiate American football athletes with a history of anterior cruciate ligament reconstruction (ACLR). This secondary analysis used data from an ongoing multicenter prospective cohort study of NCAA Division-1 American football athletes. Sprint biomechanics were collected using wearable inertial measurement units during on-field, maximal effort overground sprints. Knee kinematic variables of interest included stance time, knee flexion angle at initial contact, peak knee

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flexion angle, and knee flexion excursion during the stance phase from 2 to 5 strides near peak sprinting speed. Linear mixed effects models were used to evaluate the main effects of limb (involved vs. uninvolved), time from surgery, and their interaction on variables of interest, adjusted for peak sprint speed. Twenty male athletes (mean age: 21.2 ± 1.3 years; time from surgery: 28.0 ± 18.5 months) were fully participating in American football without limitations at the time of testing and met inclusion criteria. No significant limb × time interactions (p = 0.3-0.60) or main effects of limb (p = 0.23-0.84) or time (p = 0.08-0.84) were observed for any of our variables of interest. Division-1 American football athletes approximately 2 years post-ACLR did not demonstrate asymmetrical knee joint kinematics or stance time during sprinting. While gait asymmetries are commonly observed after ACLR during submaximal walking and running, these differences may diminish at high effort levels required during sprinting. IMU-based methods provide a clinically feasible approach for assessing on-field sprint biomechanics in athletes after ACLR.

Force and power testing during anterior cruciate ligament reconstruction rehabilitation: A world-wide survey of current practices. Sports Medicine, Ahead of Print: 1-18, 2025.

Background: During rehabilitation, the importance of restoring strength and power in a stage-based framework highlights the paramount role of testing and monitoring. Despite theoretical understanding of an optimal recovery pathway, it is unclear and inconsistent as to how practitioners implement force and power assessment following anterior cruciate ligament reconstruction (ACLR).

Objectives: We aimed to examine current worldwide practices with evaluating patients post-ACLR, identify the relative utilisation of different devices and methodologies across the rehabilitation process, and to explore the interactions between the implemented devices and respondents' self-perceived testing quality (evaluated by participants via a Likert scale).

Methods: Data were collected via an international online survey composed of 100 items, organised across 12 sections, exploring the demographics of respondents and implemented testing devices to assess individuals after ACLR and their perceived testing quality.

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Results: A total of 1154 practitioners from 78 different countries completed the survey. According to the pre-defined eight categories, 157 different combinations were recorded among practitioners. Respondents tended to use multiple devices (95.8%), with a mean of 3.6 ± 1.4. Patient assessments were most often repeated longitudinally throughout the recovery process post-ACLR (96.4%). Specific devices were used as part of "criteria-based" testing by 46.4% of respondents, "criteria- and time-based" testing by 30.8% and solely "time-based" testing by 21.9%. A significant but weak direct correlation was observed between the number of implemented devices and self-perceived testing quality ($\rho = 0.32$, $p < 0.001$), with force plates, an isokinetic dynamometer, Nordic hamstrings device and a hand-held dynamometer significantly associated with increased self-perceived testing quality ($R^2 = 0.21$, $p < 0.001$).

Conclusions: A high degree of variability in test device implementation existed among practitioners. According to the eight pre-defined device categories, over 150 different device combinations were recorded among respondents. Device use was different throughout the stages of rehabilitation and testing was primarily performed as criteria to advance patients throughout the recovery process. While acknowledging that these findings may be influenced by self-serving bias, they suggest that practitioners involved in force and power testing post-ACLR may benefit from implementing a wide range of devices, including more quantitative and objective instruments such as force plates and the isokinetic dynamometer, as these appeared to be related to higher levels of self-perceived testing quality.

Game-day temperatures are predictive of National Football League game outcomes when teams from different climates compete against each other. Temperature, Ahead of Print: 1-9, 2025.

Abstract - NFL teams may have an advantage over their opposition when players are more familiar with game-day temperatures; however, relationships between game-day air temperature and team performance had not been examined using official at-stadia air temperatures. Game data from 2017 to 25 were collected from NFL “Gamebooks.” Data were extracted for games contested by a team based in a cool climate >39° N (North teams, $n = 17$) against a team based in a warm climate <39° N (South teams, $n = 15$). Data extracted included game-day temperature (°C), game location (North, South), game outcome (win, no-win [relative to North team]), and point differential (points [relative to North team]).

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Mixed effects logistic regressions examined whether game-day temperature associated with game outcome, while mixed effects models examined whether temperature associated with point differential. Models were run for all games combined, and for games contested in North and South locations, respectively. North team win probability declined as game-day temperature increased (OR = 0.974, 95% CI [0.951, 0.997], p = .026), and North team point differential was negatively associated with temperature (β = -0.175, 95% CI [−0.286, −0.063], p = .002). Models run according to game location revealed that North team point differential was associated with game-day temperature for games played South (β = -0.152, 95% CI [−0.242, −0.063], p = .001). In NFL games contested by North (>39° N) versus South (<39° N) teams, North teams are less likely to win and have worse point differentials, as game-day temperatures increase. Research should explore whether behavioral or psychological adaptations to familiar game-day temperatures explain these findings.

Hamstring training and football performance: A systematic review and meta-analysis. International Journal of Sports Medicine, Ahead of Print: 1-50, 2026.

Abstract - The aim of this review was to evaluate the impact of strength training on hamstring muscle strength and physical performance in football players. A systematic review was conducted following the PRISMA guidelines, and the protocol was registered in PROSPERO.A literature search was performed using four electronic databases: Web of Science, SCOPUS, SportDiscuss, and PubMed. Randomized controlled trials conducted with football players aged 15 years or older were included. The studies were grouped according to moderating variables such as training protocol and physical performance outcomes. A total of 20 studies met the inclusion criteria for the review, with 10 studies included in the meta-analysis. Strength training programs demonstrated moderate to large effects on both concentric and eccentric strength of the hamstrings and quadriceps, as well as on physical performance measures such as jumping and sprinting. Eccentric overload training produced the most significant benefits in sprint and jump performance. Strength training, particularly eccentric training, significantly improves both hamstring muscle strength and physical performance in football players of male football players aged 15 years and older, highlighting its importance for injury prevention and sports performance enhancement. However, additional research is needed to evaluate whether similar benefits can be observed in female players and younger age groups.

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High rates of return to American football after primary hip arthroscopy for femoroacetabular impingement: A 2-year minimum follow-up. The Orthopaedic Journal of Sports Medicine, 14(1): 1-8, 2026.

Background: Femoroacetabular impingement (FAI) is a common cause of hip pain in young, active athletes. American football players place high demands on the hip due to the sport's requirements for pivoting, cutting, and loading in deep flexion. Although arthroscopic treatment of FAI in professional National Football League (NFL) players has shown favorable outcomes, football-specific outcomes in non-NFL athletes remain limited.

Purpose: To report the return-to-sport (RTS) rates and functional outcomes at a minimum 2-year follow-up for active American football players competing at multiple levels after primary hip arthroscopy for FAI.

Study design: Case series; Level of evidence, 4.

Methods: An institutional database was queried to identify all active American football players who underwent primary hip arthroscopy with labral repair between 2010 and 2023. Demographic characteristics, radiographic parameters, procedure details, complications, and reoperations were collected. Follow-up was completed at a minimum of 2 years to obtain updated patient-reported outcome measures (PROMs) as well as football-specific data, including current participation and RTS status. PROMs were reported as mean ± standard deviation, and categorical variables, including RTS, were reported as frequencies. Pre- versus postoperative PROMs were compared using *t* tests or Mann-Whitney *U* tests to assess outcomes.

Results: In total, 49 hips in 38 male football players (age, 18.1 years; range, 14-26 years) were included. Most athletes played at the high school (74%) or college level (21%). A total of 51% played as linemen. Labral repair was performed in all cases. At a mean 7.6 years, mean modified Harris Hip Score was 90 ± 13, Hip Outcome Score-Activities of Daily Living was 95 ± 7, and Hip Outcome Score-Sport was 89 ± 17. Overall mean surgery satisfaction was 8.7 ± 1.7 (out of 10). Significant postoperative improvements were observed in all PROMs ($P \leq .002$). Twelve players (32%) did not attempt RTS due to other nonmedical factors and were excluded from return to sport analysis. A total of 23 (88%) players achieved RTS, and 3 players (12%) did not. Two players (8%) cited other injuries, and 1 player (4%)

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continued to have hip limitations. Four hips (8%) underwent reoperation at a mean 4.9 years postoperative.

Conclusion: Amateur American football players undergoing primary hip arthroscopy for FAI demonstrated excellent patient-reported outcomes and a high RTS rate of 88%, when excluding those who did not return for nonmedical reasons, at a mean 7.6-year follow-up. Notable positional differences were observed across multiple levels of play. Although outcomes were generally favorable, both medical and nonmedical factors could limit return to sport, particularly among high school and college athletes.

Improved post-exercise rehydration with a milk permeate-based sports drink. International Journal of Sport Nutrition and Exercise Metabolism, 36: 30-41, 2026.

Abstract - This study evaluated the rehydration efficacy of a commercially available milk permeate-containing sports drink with high osmolality (MPSD) compared with a traditional sports drink (TSD), a high-potassium sports drink (HKSD), and water (W). Twelve young, healthy adults (four women) underwent a 2-hr exercise-induced dehydration protocol in the heat (29.9 °C ± 1.5 °C) until losing 2% of body weight, followed by a 4-hr rehydration period. Participants consumed on four separate occasions one of the following beverages: (a) TSD (Na+: 20 mmol/L, K+: 3.6 mmol/L, osmolality: 385 mmol/kg); (b) HKSD (Na+: 3.6 mmol/L, K+: 37.9 mmol/L, osmolality: 364 mmol/kg); (c) MPSD (Na+: 21 mmol/L, K+: 28.7 mmol/L, osmolality: 576 mmol/kg); or (d) plain water (W). Fluids (150% of body weight loss) were administered in four doses every 15 min during the first hour. Whole-body net fluid balance remained negative, but MPSD resulted in significantly higher net fluid balance at 180 min (-0.38% ± 0.3%) and 240 min (-0.66% ± 0.3%) than in other trials ($p < .05$). MPSD also led to lower cumulative urine volume (1,268 ± 173 ml) than HKSD (1,796 ± 459 ml), TSD (1,493 ± 211 ml), and W (1,565 ± 501 ml; $p < .05$). Osmotic excretion in MPSD was significantly higher than in all trials at 120, 180, and 240 min ($p < .05$), and free water clearance was lower at 180 and 240 min ($p < .05$). TSD promoted greater plasma volume restoration (1.5% ± 5%) than MPSD (-3.7% ± 6.5%) and HKSD (-5.5% ± 7.8%) at 60 min ($p < .05$). A milk permeate-based sports drink is more effective for postexercise rehydration than TSD, HKSD, and water when consumed at 150% of the body weight lost.

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Injectable testosterone replacement therapy within 1 year before ACL reconstruction is associated with increased revision rates. The Orthopaedic Journal of Sports Medicine, 15(1): 1-8, 2026.

Background: It has been demonstrated that exogenous testosterone can lead to higher rates of tendinous rupture and that injectable testosterone replacement therapy (TRT) is associated with more systemic complications. While this is the case, there are few studies exploring associations between injectable TRT and complications after anterior cruciate ligament reconstruction (ACLR) surgery.

Purpose/hypothesis: The purpose of this study is to evaluate the association between preoperative injectable TRT and ACLR revision surgery. It was hypothesize that ACLR revision surgery rates will be higher in patients undergoing TRT preoperatively compared with a matched cohort.

Study design: Cohort study; Level of evidence, 3.

Methods: A retrospective cohort of patients who underwent ACLR between 2015 and 2022 was obtained through the PearlDiver Database (Mariner). Patients with a history of ACLR, 2 years of follow-up, and injectable TRT use before ACLR were identified using International Classification of Diseases, 10th Revision, and Current Procedural Terminology codes. Three TRT usage intervals were analyzed to evaluate ACLR revision rates in those with usage of injectable TRT compared with those with propensity score-matched demographics and Charlson Comorbidity Index with no history of injectable TRT use who underwent ACLR. Cohorts were via univariate analysis, chi-square tests, Student *t* tests, and log-rank tests.

Results: TRT use ≤1 year before primary ACLR surgery demonstrated a 17.05% (66/387; mean age, 40.3) ACLR revision rate compared with 5.17% (20/387; mean age, 40.8) in the matched cohort control group ($P < .001$), yielding a relative risk of revision surgery of 3.30 (95% CI, 2.04-5.33). Statistical analysis for distant TRT use >1 year before surgery (n = 160) was underpowered but supported trends toward increased revision risk with TRT exposure with a 14.38% revision rate (n = 23). TRT users with any exposure before or after surgery (n = 2614) had similar revision rates (n = 168; 6.36%) compared with the control group (n = 164; 6.21%); the difference was not significant ($P = .82$).

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Conclusion: Our study demonstrated that injectable TRT use within 1 year of primary ACLR is associated with a >3-fold increase in ACLR revision rates with 2 years of follow-up. Physicians should consider incorporating this increased risk into patient counseling during shared decision-making discussions, as this correlation may influence clinical and surgical decision making for patients with previous ACLR.

Irrational decision-making and herd behavior: Evidence from the National Football League head coach labor market. American Behavioral Scientist, Ahead of Print: 1-11, 2025.

Abstract - We contribute to the herd behavior and behavioral economics literature by comparing the determinants of head coach candidate interviews and hires with the determinants of successful head coaches. We use National Football League (NFL) data from the 2015 to 2016 through 2021 to 2022 NFL seasons. This results in 435 candidate observations for our Interview and Hired models, as well as 582 head coaches in our Team Win Percentage and Cover Against the Spread models. Our results indicate misalignments between hiring practices and actual determinants of effective coaching. We also suggest that this irrational decision-making is associated with league herd behavior. Overall, NFL head coaches serve as the executive of the team, which allows us to provide insight into herd behavior within the executive labor market in professional sports leagues and traditional for-profit contexts.

Mental fatigue negatively impacts cognitive and physical performance outcomes: A large-scale randomized crossover trial. Medicine and Science in Sports and Exercise, 58(2): 225-241, 2026.

Purpose: Although many studies have demonstrated that prolonged cognitive load can impair subsequent cognitive and physical performance, commonly described as mental fatigue, more recent findings suggest that the relationship is more nuanced. Moreover, significant heterogeneity between different smaller studies makes it difficult to summarize definitive conclusions. The aim of the present study was to provide a more robust examination of the detrimental effects of sustained cognitive

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demands on cognitive and physical performance fatigability using a large sample within a controlled environment.

Methods: One hundred seventeen participants (57 female; 32 ± 9 yr) were included in this randomized counterbalanced crossover experiment (NCT05576935) consisting of familiarization, intervention (45-min individualized Stroop task), and control (45-min self-chosen documentary) sessions. Performance was evaluated using a 6-min GoNoGo task and a 20-min cycling time trial. Multiple secondary outcomes, such as rate of perceived exertion (RPE), feelings of fatigue, and motivation, were incorporated to explore their relationship to possible behavioral effects.

Results: There was a significant worsening of GoNoGo reaction time between conditions ($P < 0.001$; $\eta p2 = 0.07$). A trend toward significance was found regarding the negative influence of the Stroop task on time trial distance ($P = 0.059$; $d = 0.20$), which was linked to a significant decrease in cadence ($P = 0.042$; $d = 0.22$) in the intervention condition. Moreover, the feelings of physical fatigue ($P < 0.001$; $\eta p2 = 0.01$), mental fatigue ($P < 0.001$; $r = 0.37$-0.47), and RPE ($P = 0.002$; $\eta p2 = 0.01$) remained higher during the time trial in the intervention condition compared with the control condition.

Conclusions: In the present study, prolonged cognitive effort impaired specific cognitive and physical performance outcomes, consistent with the characteristics of mental fatigue. Secondary outcomes show an important role for the feeling of fatigue in the determination of physical performance.

Muscle performance but not biomechanics associate with second knee injury in a matched cohort of athletes who passed functional return-to-sport criteria after ACL reconstruction. Knee Surgery, Sprots Traumatology, Arthroscopy, Ahead of Print: 1-13, 2025.

Purpose: Athletes remain at risk for a second knee injury after anterior cruciate ligament reconstruction (ACLR), even after passing return-to-sport (RTS) testing. While biomechanical asymmetries have been linked to reinjury, it is unclear whether deficits persist in athletes who meet RTS criteria. This study compared muscle performance and biomechanical function between athletes who sustained a second knee injury after passing RTS testing and matched controls without reinjury.

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Methods: In this case-control study, 11 athletes who returned to their preinjury sport within one year after ACLR, passed RTS testing, and subsequently sustained a second knee injury were matched to controls without reinjury. Muscle performance and biomechanical assessments were conducted within two weeks of passing RTS testing, prior to second knee injuries. Muscle performance was assessed using isometric and isokinetic strength testing on an electromechanical dynamometer. Biomechanical function was evaluated during bilateral drop vertical jumps, unilateral drop vertical jumps, and single-legged hop for distance using three-dimensional motion analysis with embedded force platforms. Muscle performance and biomechanical function were compared between limb (injured, uninjured) and group (second knee injury, no second knee injury) using 2 × 2 mixed analyses of variance (ANOVA).

Results: Eight athletes reinjured the ipsilateral knee and three injured the contralateral knee, with reinjury occurring on average 15.4 ± 5.0 months after ACLR. Compared with controls, the second injury group showed greater quadriceps rate of torque development symmetry during early isometric contraction (0-100 ms; $p = 0.008$). No other significant differences in muscle performance or biomechanical variables were observed.

Conclusion: Greater and more symmetric quadriceps RTD was associated with second knee injury, while biomechanical measures did not differ between groups. These findings suggest that conventional RTS tests may not adequately detect at-risk athletes after ACLR, and broader assessment strategies may be needed to guide targeted injury prevention.

Level of evidence: Level III.

Physiological adaptations and practical efficacy of different blood flow restriction resistance training modes in athletic populations. Frontiers in Physiology, 16: 1683442, 2025.

Abstract - Blood flow restriction resistance training enhances athletic adaptations via distinct mechano-metabolic pathways. This review synthesizes evidence comparing three blood flow restriction resistance training modalities: Low-load resistance training with blood flow restriction (using 20%-30% of one-repetition maximum) prioritizes metabolic stress (lactate and hydrogen ion accumulation, cellular

swelling), activating growth hormone (GH)/insulin-like growth factor 1 (IGF-1)/mechanistic target of rapamycin (mTOR) pathways to promote type I muscle fiber hypertrophy, making it suitable for joint-sparing rehabilitation scenarios. Supplemental blood flow restriction resistance training programs combine high-load tension (utilizing 75%-90% of one-repetition maximum) with additional blood flow restriction to produce an acute synergistic effect. This method enhances the recruitment of type IIa/x muscle fibers and prolongs mTOR phosphorylation. Combined blood flow restriction resistance training employs alternating cycles of high-load phases (70%-85% 1RM) and blood flow restriction phases (hypoxia-inducible factor 1-alpha (HIF-1α)-mediated angiogenesis), optimizing phosphocreatine resynthesis and neural drive to achieve specialization of type IIx muscle fibers. Periodized application requires matching modalities with training phases: combined blood flow restriction training for hypertrophy during the preparatory phase, supplemental blood flow restriction training for strength maintenance during the competitive phase, and low-load resistance training with blood flow restriction for active recovery. This mechanistic framework supports evidence-based blood flow restriction resistance training programming to maximize athletic adaptations while mitigating injury risk.

Positional comparison of countermovement jump performance in American collegiate football athletes. International Journal of Sports Science & Coaching, Ahead of Print: 1-8, 2026.

Purpose: The purpose of this investigation was to examine countermovement jump differences between positional groups in American football.

Methods: 85 Division I collegiate football players were assigned to one of three groups (lineman =32, midskill =29, skill =24). Each participant performed three CMJ trials using a portable force platform. A one-way analysis of variance was performed on all variables of interest. An a priori significance level of $p < 0.05$ was used for all analysis.

Results: Significant differences were present across all three groups for jump height ($p < 0.01$) and RSImod ($p < 0.001$) with the skill group having the highest values and the linemen having the lowest. Inversely, take-off momentum and body mass were the greatest in the linemen group with the lowest

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being the skill group ($p < 0.001$). Linemen had longer duration in the propulsive phase ($p < 0.01$), braking phase ($p < 0.01$), and time to take off ($p < 0.01$).

Conclusions: The differences between groups can aid in the design of training programs targeting specific traits for each positional group. Additionally, differences in positional metrics of interest may exist when using the CMJ to monitor fatigue status over the course of a competitive season.

Science or hype? Non-invasive brain stimulation and sport-specific performance: Evidence from a systematic review and three-level meta-analysis. Journal of Sports Sciences, Ahead of Print: 1-25, 2026.

Abstract - Enhancing athletic performance is a core objective in competitive sports. Non-invasive brain stimulation (NIBS), a set of neuromodulatory techniques, aims to modulate cortical excitability and thereby influence motor and cognitive functions. Despite its commercial appeal and growing application in sports contexts, the empirical evidence supporting its efficacy remains inconclusive. A three-level meta-analysis was conducted to evaluate the effects of NIBS on sport-specific performance and to identify potential moderating factors. The present study included 25 eligible studies and 64 independent effect sizes. Notably, all included studies employed transcranial direct current stimulation (tDCS). The meta-analysis revealed a small but significant overall effect of tDCS on sport-specific performance (Hedges' $g = 0.35$, $p < .01$), with considerable heterogeneity across studies ($I^2 = 59.13\%$). tDCS appeared more effective for open-skill sports ($g = 0.63$, $p < .01$) and yielded benefits when targeting motor executive regions ($g = 0.56$, $p < .01$). The overall quality of evidence was rated as low. Therefore, while the findings suggest that tDCS may enhance certain aspects of sport-specific performance, the evidence base for NIBS as a whole remains preliminary and methodologically limited. Consequently, the purported performance benefits of NIBS should be interpreted with caution by both researchers and practitioners.

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Serial ultrasound imaging leading up to an Achilles tendon rupture after identification of a thickened paratenon: A case report. American Journal of Physical Medicine & Rehabilitation, 104(1): 1067-1069, 2025.

Abstract - A 22-yr-old lacrosse student-athlete had participated in a research study examining the Achilles tendons of asymptomatic athletes, with normal appearance of his Achilles tendons bilaterally 7 mos before injury. Around 4 wks before injury, the patient started to report achy pain in the affected Achilles tendon, exacerbated by ankle dorsiflexion and plantarflexion, in the context of using new cleats during conditioning. He underwent repeat ultrasound assessment, identifying a normal appearing Achilles tendon, though with new hypoechogenic paratenon thickening approximately 1-4 cm proximal to the calcaneus. He ultimately suffered a noncontact complete Achilles tendon rupture. This case is the second known example of an Achilles tendon rupture with previous asymptomatic advanced imaging and additionally calls into question the role of the Achilles paratenon in Achilles tendon pathology and rupture.

Short-term change of physical capacities of football players within few days before ACL injury: A retrospective case-control study. BMC Sports Science, Medicine, and Rehabilitation, Ahead of Print: 1-16, 2026.

Purpose: Anterior cruciate ligament (ACL) ruptures represent one of the most severe injuries in professional football, often resulting in long rehabilitation, impaired performance, and increased risk of re-injury. The aim of this study was to investigate whether performance parameters derived from match statistics can serve as early indicators of ACL rupture in professional male football players.

Methods: A retrospective case-control design was applied. Forty-two male professional football players from the German Bundesliga and 2. Bundesliga with confirmed ACL ruptures between 2016 and 2024 were included, alongside 42 matched controls from the same teams and positions. Match performance data from ten games preceding the injury were analyzed. Parameters included minutes played, total distance covered, number of sprints, maximal speed, pass accuracy, number of duels, and duel success rate. Independent t-tests compared injured and control players across individual matchdays and

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aggregated intervals (the average values across the last four, three, and two matches before injury). Additionally, odds ratios (OR) with 95% confidence intervals were computed based on upper (≥ 75th percentile) and lower (≤ 25th percentile) quartile thresholds to quantify the relative risk associated with extreme performance values.

Results: Injured players showed higher maximum speed that consistently differentiated them from controls, with significant differences at matchday 2 ($p = 0.005$, OR = 3.42, 95% CI 1.45-8.06) and across all aggregated intervals ($p = 0.015$-0.031). Injured players also showed significantly fewer minutes played at matchday 2 before injury ($p = 0.046$, OR = 2.36, 95% CI 1.01-5.51) and across certain intervals (last four and three matches before injury; $p = 0.027$-0.044). Analysis of matchdays 5-10 revealed no significant group differences for any performance parameter, confirming that relevant performance changes manifest primarily in the immediate pre-injury period. No significant group differences emerged for distance covered, sprint count, pass accuracy, or duel frequency between ACL injured players and controls.

Conclusion: Maximum speed showed the strongest association with ACL rupture risk, with significant differences at matchday 2 ($p = 0.005$) and across aggregated intervals (the average values across the last four, three, and two matches before injury). Reduced playing time emerged as an additional indicator. Although distance covered, sprint count, and pass accuracy did not reach statistical significance individually, their temporal patterns revealed a predisposing risk constellation: injured players demonstrated reduced cumulative exposure combined with acute high-intensity spikes at matchday 2, declining technical precision, and increased physical confrontation at matchday 1. This suggests ACL injury risk manifests through deterioration of integrated performance capacity under acute load fluctuations rather than isolated thresholds. Multifactorial approaches integrating biomechanical, physiological, and temporal performance patterns are essential for effective ACL injury prevention in professional football.

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Sleep and Athletic Performance: A multidimensional review of physiological and molecular mechanisms. Journal of Clinical Medicine, 14: 7606, 2025.

Abstract - Sleep is a fundamental biological process in athletes, indispensable for tissue regeneration, exercise adaptation, and injury prevention. Disruptions in sleep architecture and duration have been consistently associated with diminished physical performance and adverse health outcomes, impairing muscular strength, power output, and endurance capacity, and concurrently compromising cognitive function. On a physiological level, insufficient sleep disrupts endocrine homeostasis, elevating cortisol levels and reducing anabolic hormones such as testosterone and growth hormone. At the molecular level, sleep loss promotes the upregulation of pro-apoptotic gene expression and exacerbates pro-inflammatory signaling pathways. Optimal sleep duration and quality represent a critical "regenerative window", essential for enhancing athletic performance and safeguarding physiological resilience. Ensuring adequate sleep among athletes can be effectively achieved through educational, behavioral, and nutritional interventions outlined in this review.

The association between early sport specialization and injury and career outcomes among National Football League athletes. European Journal of Sport Science, 26: e70120, 2026.

Abstract - Early sport specialization during adolescence has been linked to overuse injuries in several sports, yet its long-term impact in American football remains underexplored. The purpose of this study is to examine whether early specialization in football during high school is associated with higher injury rates and shorter careers than multi-sport participation. This is a retrospective cohort study with level of evidence 3. We analyzed all NFL players drafted from 2011 to 2023 (n = 2556) who played ≥ 16 career games. Athletes were classified as multisport or single-sport based on high school varsity participation using public records. Injury data were obtained from validated online databases. The primary outcome was injury incidence, measured as injuries per 1000 snaps (defined as individual plays participated in). Secondary outcomes included career length, total games played, and weighted career approximate value (AV). Injury rates were compared with position-stratified incidence rate ratios (IRRs). Multisport athletes (63.6%) sustained fewer total injuries (IRR, 0.80, 95% CI, 0.76-0.85, and $p < 0.001$) and major

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injuries (IRR, 0.77, 95% CI, 0.71-0.82, and $p < 0.001$) compared to single-sport athletes. Multisport athletes also played 12.2 more NFL games (95% CI, 9.2-15.1, $p < 0.001$, and $d = 0.32$) and had 0.7 additional career years (95% CI, 0.5-0.9, $p < 0.01$ and $d = 0.28$). NFL players who participated in multiple sports during high school had significantly lower injury rates and greater career durability. These findings support the body of evidence discouraging early sport specialization.

The Power of Belief: Investigating the placebo effect in post-exercise recovery strategies for football players. Healthcare, 14: 4, 2026.

Objectives: The objective is assessing whether the placebo effect can influence the time course of recovery following a football match.

Methods: Using a randomized crossover design, eighteen youth male players (age: 15.3 ± 0.5 years, stature: 178.7 ± 6.4 cm, body mass: 65.3 ± 7.6 kg, playing experience: 8.6 ± 1.5 years) completed two friendly matches, followed by placebo (PLA; sham vagus nerve stimulation) or passive rest (CON). To assess the impact of PLA, countermovement jump height (CMJ), 10 and 20 m sprint times, heart rate variability (Ln-rMSSD), static and dynamic muscle soreness, and perceived fatigue were measured at pre-match, post-match, post-recovery, and 24 h post-match.

Results: Our findings indicate that match play induced substantial fatigue, with significant deteriorations [$p \leq 0.002$, *small-to-large* effect sizes (ES)] in CMJ, 10 and 20 m sprint performance, Ln-rMSSD, muscle soreness, and perceived fatigue at post-match and post-recovery compared to pre-match (except Ln-rMSSD between pre-match and post-recovery: $p = 0.151$, *small* ES). Although no significant between-intervention differences were found for any variable at any time point ($p > 0.05$), effect size analysis showed *moderately* lower perceived fatigue ($r = 0.40$) and dynamic soreness ($r = 0.32$) in PLA compared to CON at post-recovery.

Conclusions: These findings suggest that while placebo stimulation did not affect performance or heart rate variability, it may support perceptual recovery. This holds relevance for both research and practice, as including placebo conditions can help isolate psychological effects from true treatment responses, while promoting positive expectations may enhance the perceived effectiveness of recovery strategies.

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Pro Football Sports Science Research Update

The protective effect of preseason running workload against in-season hamstring strain injuries in elite soccer players. Sports Health, Ahead of Print: 1-7, 2025.

Background: Preseason training plays a pivotal role in preparing athletes for the competitive season and reducing the risk of hamstring strain injuries (HSI). This study evaluated the protective effect of preseason running workload on the risk of in-season HSI in elite soccer players.

Hypothesis: Higher preseason sprint-related metric is associated with reduced risk of in-season HSI in elite soccer players.

Study design: Cohort study.

Level of evidence: Level 3.

Methods: Global positioning system data and HSI records from 67 elite Japanese soccer players between 2021 and 2023 were analyzed. Variables included internal workload, total distance, high-speed distance, sprint distance, sprint count, and acceleration/deceleration count. Multivariable logistic regression analysis determined the association between workload metrics and HSI risk.

Results: A total of 17 HSI incidents were recorded during the study period. Increased preseason sprint distance was associated significantly with a reduced risk of in-season HSI (odds ratio [OR], 0.896; 95% CI, 0.822-0.976; $P = 0.01$), even after adjustment (model 1: OR, 0.892; 95% CI, 0.814-0.979; $P = 0.02$, model 2: OR, 0.899; 95% CI, 0.822-0.983; $P = 0.02$). Higher sprint count was associated with decreased HSI risk (OR, 0.761; 95% CI, 0.633-0.916; $P = 0.004$), with consistent findings after adjustments (model 1: OR, 0.755; 95% CI, 0.623-0.916; $P = 0.004$, model 2: OR, 0.763; 95% CI, 0.631-0.923; $P = 0.005$). The area under the curve values for sprint distance and sprint count were 0.721 and 0.739, respectively.

Conclusion: Preseason training marked by higher sprint distances and counts demonstrates a protective effect against in-season HSI in elite soccer players, supporting comprehensive preseason running workload as an essential strategy for reducing injury risk and enhancing season readiness.

Clinical relevance: Incorporating sprint-focused training during the preseason may reduce HSI incidence and minimize injury-related absences in elite soccer players.

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Pro Football Sports Science Research Update

The use of fantasy points to evaluate return-to-play performance after time-loss injuries in the National Football League. Cureus, 17(12): e99333, 2025.

Background: National Football League (NFL) "fantasy leagues," in which fans accumulate points based on the NFL players' in-game performance, have boomed in popularity. There is a paucity of objective measures to track player performance following injuries. Therefore, our study evaluates whether fantasy football points can serve as an assessment tool for player recovery after time-loss injuries.

Methods: In this descriptive epidemiology study, injuries in offensive skill positions over five NFL seasons (2017-2022) resulting in at least one missed game were retrospectively reviewed. Return to performance was assessed by the difference in fantasy points per game (PPG) between the injury year and the following season.

Results: Reported injuries resulted in a change of -0.50 (CI: -0.67, -0.33) PPG. Quarterbacks (QBs) had the greatest decrease in PPG at -1.95 PPG (CI: -3.47, -0.44) followed by WR at -0.33 PPG (CI: -0.53, -0.12). Injury to the shin resulted in the greatest decrease in PPG at -3.24 PPG (CI: -7.69, 1.2). Injury to the arm had the second greatest decrease in PPG at -2.65 PPG (CI: -6.78, 0.04). No association between the number of games missed and difference in PPG was found (R^2=0.0047). Injury prior to a bye week did not alter PPG, yet injury after a bye week lead to increased loss of -1.22 PPG (CI: -1.491, -0.964). Players in the league for three years or more experienced a decline in PPG recorded of -17.10 (CI: -21.61, -12.591).

Conclusion: The use of fantasy football points may serve as an objective tool for assessing player return to performance post-injury, though correlation with rigorous clinical evaluation is still needed, especially since fantasy points lack the precision needed for injury severity and since various team-related factors may influence fantasy point performance. QBs and players with shin or arm injuries experience the greatest decline in fantasy PPG. Longer NFL careers may also contribute to performance decline, while injury timing has a lesser effect.

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Pro Football Sports Science Research Update

Transdermal uptake of substances banned in sports and its relevance for doping controls. International Journal of Sports Medicine, Ahead of Print: 1-19, 2025.

Abstract - Transdermal drug delivery has emerged as an area of significant interest in both pharmaceutical sciences and antidoping research. This narrative review explores the implications of transdermal uptake of substances banned in sports, emphasizing its relevance for doping control analysis. The human skin's complex role as both a barrier and a site of drug absorption is examined, alongside studies investigating the transdermal administration of substances on the World Anti-Doping Agency Prohibited List, including approved transdermal formulations and drug candidates. Recent cases demonstrate that (unintentional) dermal exposure of athletes to substances banned in professional sports, e.g., clostebol, can lead to an adverse analytical finding in doping controls. Consequently, studies investigating the detectability and metabolism of transdermally administered doping substances, which might even support identifying the route of exposure, are becoming increasingly important. This review emphasizes the relevance of addressing the skin as a pathway for both intentional doping and unintentional contamination, outlining critical implications for global antidoping efforts and future research priorities.

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